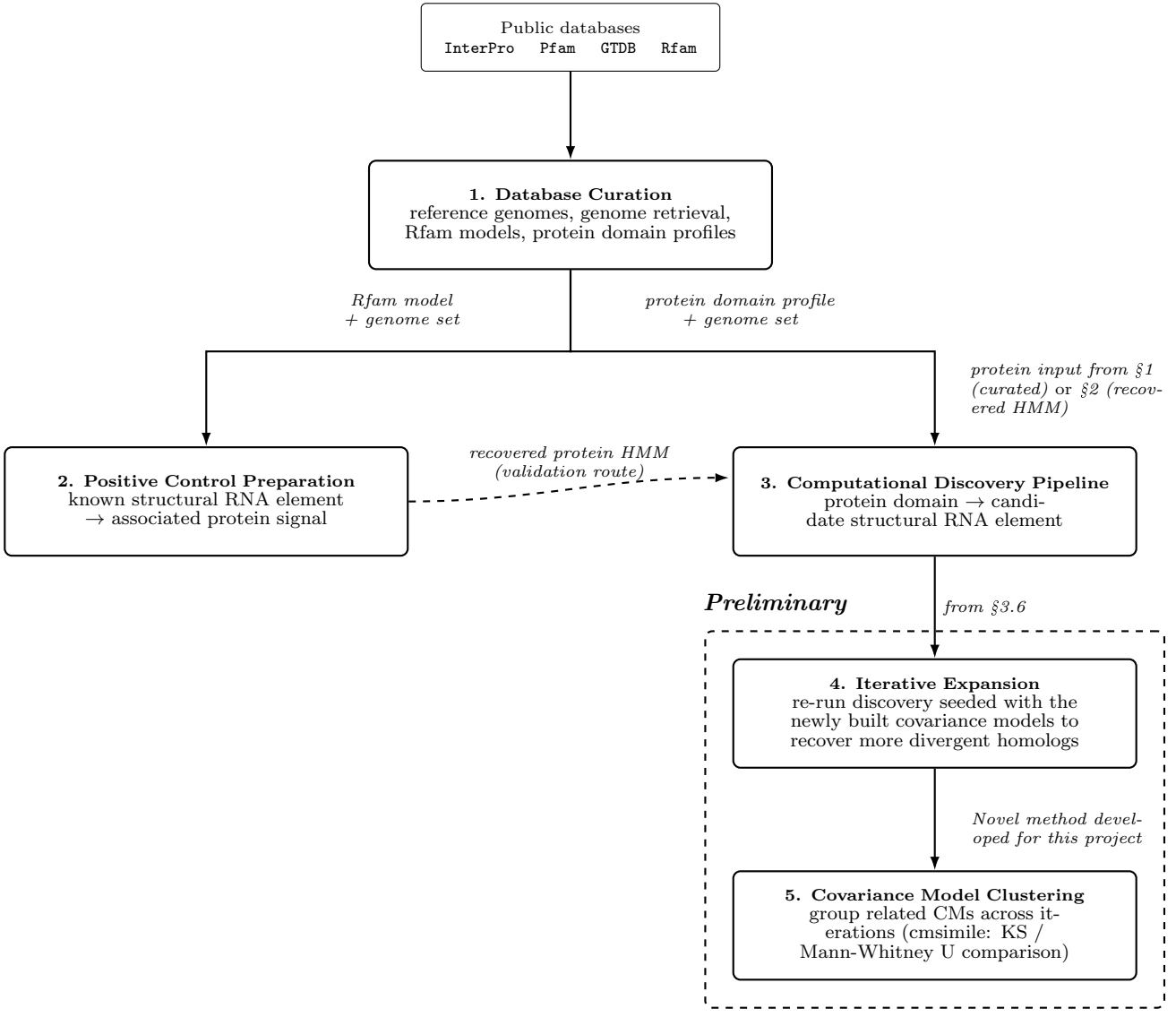

Public databases
InterPro Pfam GTDB Rfam
1. Database Curation
reference genomes, genome retrieval, Rfam models, protein domain profiles
Rfam model + genome set
protein domain profile + genome set
protein input from §1 (curated) or §2 (recovered HMM)
2. Positive Control Preparation
known structural RNA element → associated protein signal
recovered protein HMM (validation route)
3. Computational Discovery Pipeline
protein domain → candidate structural RNA element
Preliminary
from §3.6
4. Iterative Expansion
re-run discovery seeded with the newly built covariance models to recover more divergent homologs
Novel method developed for this project
5. Covariance Model Clustering
group related CMs across iterations (cmsimile: KS / Mann-Whitney U comparison)